CARISURG MEDTECH PATHWAYS RESEARCH PROJECT

PRELIMINARY PROPOSAL

Project Title: An Explainable AI Assistant to Support Emergency Department Triage in the Caribbean

Student Name: Sariana Ramoutar

Statement of the Problem:

Emergency departments rely on rapid triage decisions to prioritise patient care, but busy clinical environments, resource constraints and human subjectivity can lead to inconsistent triage outcomes (De Freitas et al., 2020; Seo et al., 2024). In recent studies, artificial intelligence has demonstrated strong potential in improving triage speed and predictive performance. However, many systems function as "black boxes," meaning they provide recommendations without explaining the reasoning behind them (Da'Costa et al., 2025; Tyler et al., 2025). As a result, clinicians may find it difficult to understand, verify and confidently incorporate AI recommendations into routine decision-making, limiting their practical use in emergency care (Bhuvaneswari et al., 2025).

Previous Work:

Recent clinical literature highlights both the potential benefits and implementation challenges of AI within emergency triage.

- Multiple studies report that machine learning models frequently can outperform traditional triage approaches and support patient prioritisation, admission prediction and resource planning (Tyler et al., 2025; Da'Costa et al., 2025; Almulihi et al., 2024). Researchers have applied these models to predict immediate Emergency Severity Index (ESI) levels, downstream hospital admissions, ICU placements, and overall patient mortality risks (Agius et al., 2025; Almulihi et al., 2024; Tschoellitsch et al., 2023). Furthermore, advancements in Natural Language Processing (NLP) show that AI can extract clinically relevant information from free-text nursing notes that may be difficult for traditional systems to analyse (Lee et al., 2024).
- Triage accuracy is influenced by factors such as nurse experience and the amount of time available for patient assessment (Seo et al., 2024). Research suggests that clinicians often use "situational triage" to balance patient risks against operational pressures such as overcrowding and limited staffing. Observational research further demonstrates that emergency department workflows are shaped by staffing shortages, resource limitations and patient flow bottlenecks, requiring clinicians

to continually adapt their practices to maintain patient movement (De Freitas et al., 2020). These findings support the development of human-centred AI frameworks that complement existing workflows rather than disrupt them (Bhuvaneswari et al., 2025; Gorick et al., 2023).

- Despite promising performance in research settings, several challenges remain before widespread clinical adoption. First, some studies report a tendency toward over-triage, which may increase pressure on limited healthcare resources (Tyler et al., 2025). Second, algorithmic bias remains a concern, particularly when models are trained on datasets that do not adequately represent diverse patient populations (Da'Costa et al., 2025; Huang et al., 2022). Finally, many published systems have limited external validation and are trained on isolated, single-institution datasets, making it difficult to determine how well they would perform in different healthcare settings (Agius et al., 2025; Araouchi & Adda, 2024; Tschoellitsch et al., 2023). This limitation is particularly important for Caribbean healthcare settings, where workflow changes and resource availability may differ substantially from environments in which most AI systems have been developed and evaluated (De Freitas et al., 2020).

Collectively, these studies indicate that while strong predictive performance is achievable, a gap remains for transparent, explainable triage tools that integrate seamlessly with existing clinical workflows while remaining appropriate for resource-constrained Caribbean emergency departments.

Proposed Solution:
This project will develop and evaluate an explainable AI decision-support prototype designed to assist clinician triage decisions using a de-identified, publicly available dataset. In contrast to traditional "black box" systems, the proposed model will simultaneously predict patient acuity levels and generate a visual explanation showing which clinical features and vital signs most influenced each recommendation.

The proposed model aims to achieve triage classification performance comparable to previously reported benchmark studies (e.g., Araouchi & Adda, 2024), while introducing explicit visual explainability features to address the trust and transparency concerns highlighted in the literature (Da'Costa et al., 2025). This prototype serves as a foundational step toward developing interpretable triage software that can be adapted to the workflow, infrastructure and resource constraints of Caribbean emergency departments.

Constraints & Stakeholders:

Successful implementation of an AI-assisted triage system depends not only on predictive performance but also on how well it fits within existing emergency departments workflows. The following workflow constraints and stakeholder concerns were identified through the literature and will guide the design of the proposed explainable AI prototype.

<u>Workflow Constraints</u>

1. Limited information available at triage

Emergency clinicians often make triage decisions before complete patient information is available. Laboratory results, previous medical records and additional clinical documentation may only become available later in the patient journey, creating downstream workflow bottlenecks (De Freitas et al., 2020). AI models must often rely on extracting meaning from highly variable, unstructured initial nursing notes instead of complete datasets (Lee et al., 2024). An AI-assisted triage system must therefore be capable of generating recommendations using only information typically available during the initial triage assessment, such as presenting complaints, vital signs and nurse observations.

2. High workload and time pressure

Emergency departments are fast-paced, highly resource-constrained environments where staffing shortages frequently occur, sometimes leaving shifts operating with less than half of their required nursing staff (De Freitas et al., 2020). Triage nurses must assess patients, document findings and manage patient flow within a limited timeframe, and triage accuracy is directly tied to the amount of evaluation time available (Seo et al., 2024). Any AI solution that requires additional data entry or complex interactions may increase workload and reduce adoption. The system should therefore integrate into existing workflows and provide recommendations with minimal additional effort.

3. Existing triage processes must be maintained

Emergency departments rely on established triage procedures to prioritise patient care and coordinate patient movement. Triage nurses frequently rely on holistic reasoning, clinical intuition and situational judgement rather than rigid, unyielding algorithms (Gorick et al., 2023). Introducing additional approval steps or workflow interruptions could delay care and reduce efficiency. The proposed AI system should function as a decision-support tool within existing triage processes rather than replacing clinician judgement or changing current workflows (Bhuvaneswari et al., 2025).

Clinical Stakeholders

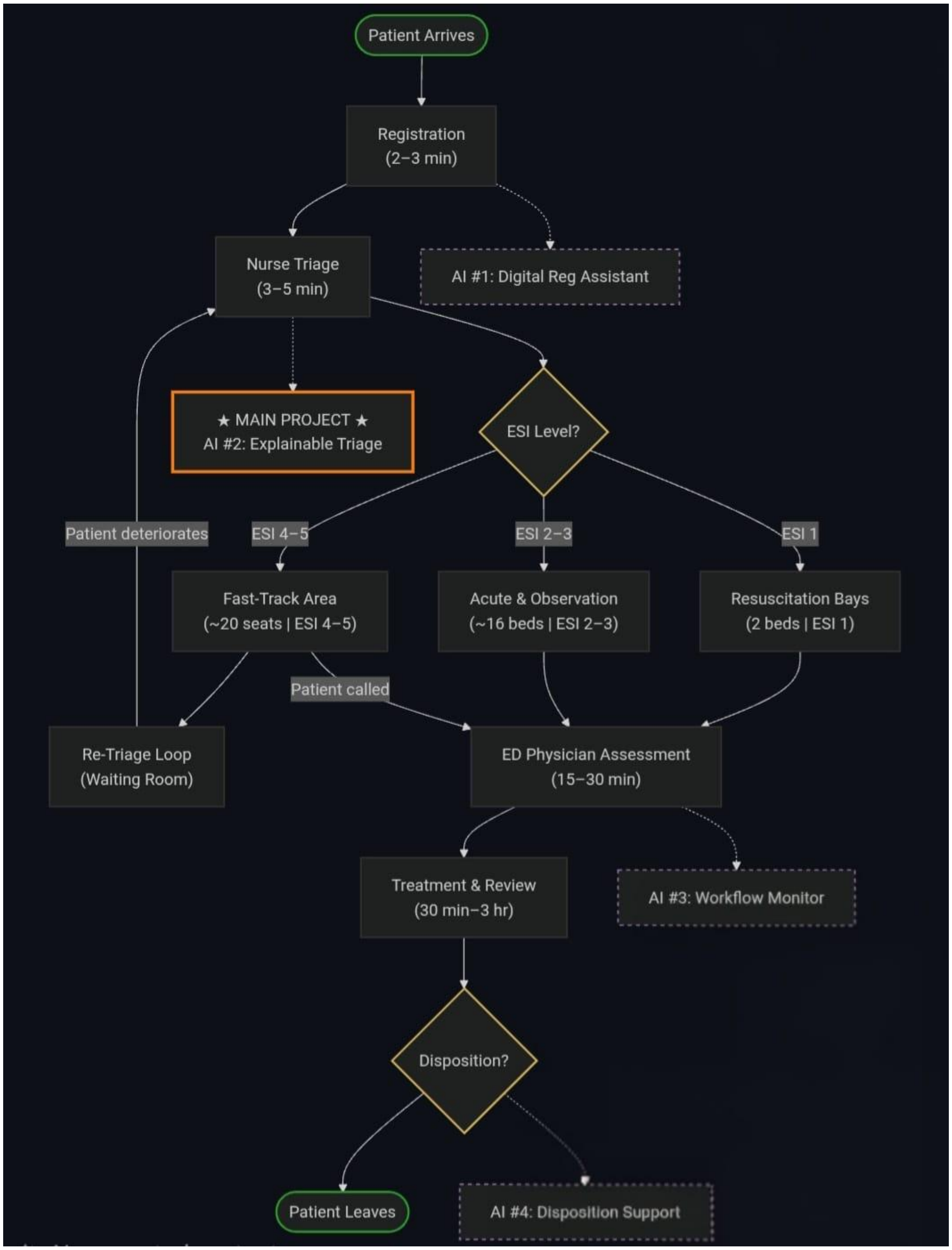
Stakeholder	Primary Concern	Supporting Literature Evidence
Triage Nurse	Recommendations must be accurate, understandable and delivered quickly enough to support real-time decision-making without increasing workload.	Requires tools that align with holistic reasoning rather than rigid rules (Gorick et al., 2023). Decision support is vital as their evaluation time directly impacts accuracy (Seo et al., 2024).
Emergency Department Consultant	AI recommendations should improve patient prioritisation while maintaining clinician oversight and patient safety.	Emphasises that AI must remain a supportive mechanism that preserves ultimate human oversight and clinical adoption (Bhuvaneswari et al., 2025; Da'Costa et al., 2025).
Administrative Clerk	The system should integrate with existing registration and electronic health record processes without creating duplicate documentation.	Registration and clerical steps are the very first bottleneck in patient flow; systems must adapt to highly localised documentation styles (De Freitas et al., 2020; Lee et al., 2024).
Porter / Support Staff	Earlier identification of high-priority patients should improve patient movement and reduce delays between care areas.	When porters are delayed, highly-trained clinicians are forced to "substitute down" to wheel patients to scans manually (De Freitas et al., 2020).
Patient / Patient Representative	AI-assisted triage should be transparent, fair and support timely access to care while ensuring clinicians remain responsible for final decisions.	Demands strict guardrails regarding algorithmic fairness, transparent explanations and equity across diverse, vulnerable patient populations (Da'Costa et al., 2025).

Mercer General Hospital Context

Mercer General Hospital is a fictitious Caribbean regional referral hospital that uses a nurse-led, five-level Emergency Severity Index (ESI) triage process. Patients are initially assessed using information available at presentation, including chief complaint, vital signs and brief clinical history, before being assigned to an appropriate treatment area. The proposed AI assistant is intended to support this initial triage stage and operate within existing workflows rather than replace clinician judgement.

Workflow Diagram showing the Current Triage Process for Mercer ED:

This diagram maps the patient path through the Mercer General Hospital Emergency Department from arrival to discharge. It traces how patients are registered, assessed by a triage nurse, and assigned to a treatment area based on their ESI score. The diagram also marks four areas where AI can improve the workflow, with this project focusing specifically on **AI Opportunity #2 (Explainable Triage)** to assist nurses during the initial assessment.



Process Notes for the Workflow Diagram

- **Registration** (2–3 min): Demographic and administrative information is collected before clinical assessment begins. AI Opportunity #1 (Digital Registration Assistant) represents a possible future solution to streamline this stage but is outside the scope of this project.
- **Nurse Triage** (3–5 min): A triage nurse performs the initial assessment using information immediately available at presentation, including the chief complaint, vital signs and brief medical history. This stage determines the patient's ESI level.
- **AI Opportunity #2 (Main Project)**: This project focuses on providing explainable decision support during the initial triage assessment. The system is intended to assist nurses by explaining the factors influencing its recommendations while preserving clinician oversight.
- **ESI-Based Patient Routing**: Patients are directed to different treatment areas according to urgency. ESI 1 patients proceed directly to the resuscitation bays, ESI 2–3 patients are managed in acute and observation areas, and ESI 4–5 patients are placed in the fast-track area.
- **Re-Triage Loop**: Patients waiting in the fast-track area may deteriorate and require reassessment. Re-triage is part of standard clinical practice and allows patients to be upgraded to higher-acuity pathways when necessary.
- **Physician Assessment** (15–30 min) **and Treatment** (½–3 hr): Emergency physicians assume responsibility for diagnosis and management. Additional investigations, treatment and specialist consultations may occur before a disposition decision is made.
- **AI Opportunities #3 and #4**: Workflow monitoring and disposition support represent additional areas where AI could be applied. However, these functions occur later in the patient journey and are outside the scope of the proposed work.
- **Project Scope**: The proposed system operates only during the triage stage and does not replace clinical judgement, modify existing workflows or eliminate the need for re-triage and ongoing patient monitoring.

Meanings of Emergency Severity Index (ESI) Score

ESI	Meaning	Target Time to be Seen
1	Life-threatening – requires immediate intervention (cardiac arrest, major trauma, shock)	Immediate
2	High-risk, severe pain, disoriented, or vitals in danger zone (chest pain, suspected stroke)	≤ 10 min
3	Needs multiple resources to stabilise; vitals not in danger zone (sepsis, abdominal pain)	≤ 30 min
4	Needs one resource (simple laceration, urinary tract infection)	≤ 60 min
5	Needs no clinical resources (prescription refill, minor complaint)	Can wait

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